

Task 7: Table Relationships & Foreign Keys

Tools:

- MySQL Workbench
- Alternatives: PostgreSQL, BigQuery Sandbox

Hints / Mini Guide:

1. Create two tables named departments and employees where departments holds department details and employees stores employee records.
2. Define a PRIMARY KEY in the departments table to uniquely identify each department.
3. Add a department_id column in the employees table that references the departments table.
4. Apply a FOREIGN KEY constraint to enforce referential integrity between the two tables.
5. Insert valid records and observe successful inserts.
6. Attempt inserting invalid foreign key values and analyze the error messages.
7. Implement ON DELETE CASCADE and understand how dependent records behave.
8. Explain real-world mapping such as company → departments → employees.

Deliverables:

- SQL script with table creation, foreign key constraints, and test insertions

Final Outcome:

- Intern understands relational database design and referential integrity enforcement.

Interview Questions Related To Above Task:

- What is a foreign key?
- Difference between primary key and foreign key?
- What is referential integrity?
- What does ON DELETE CASCADE do?
- Can a foreign key accept NULL values?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

